

LOT SYSTEMS CORPORATION

COSMO® Cube

Overview & Components Buying List

A hardware computer connected to lot-systems.com

Document v1.0 · Recovered and shipped 2026-09-03
Originally designed 2026-06-12

Inventor: Vadik Marmeladov — COSMO® CIA

Device Identity

Field	Value
Product name	COSMO® Cube
Form factor	40mm × 40mm × 5mm flat square
Body	316L stainless steel, CNC machined, 2 parts
Side A (back)	Mirror-polished #8 finish — LOT® engraved
Side B (front)	Satin SS — OLED display, camera, Copy button
Weight	~28g
Connectivity	WiFi 802.11n, BLE 5.0
Charging	Qi wireless, 5W, pad included
Platform	lot-systems.com
Production run	100 units
Unit cost / retail	~\$110 cost / \$349 retail
Manufacturer	PCBWay (PCB + SMT + CNC)

Core Electronics

Function	Part	Why
MCU	ESP32-S3-MINI-1U	Native WiFi + BLE, on-device AI vector accel
Display	SSD1327 1.0" OLED, 128×128	Square format, 16-level grayscale, thin
Camera	Himax HM01B0	1.1mW at 30fps, CSP 2.45mm package
Weather	Bosch BME280	Temp / humidity / pressure, industry standard
IMU	TDK ICM-42688-P	Motion, orientation, tap-to-wake
Gesture / light	Broadcom APDS-9960	Wake-on-approach, dismiss gesture
Qi receiver	TI BQ51013B	5W wireless charging
PMIC	TI BQ25892	Charge + system rail management
Battery	280mAh LiPo, 35×35×2.5mm	~18h active, ~4 days standby

Components Buying List — 100-Unit Run

Full sourcing detail (MPNs, backup suppliers, datasheets) is in COSMO-BOM-v1.md. This is the summary for procurement sign-off.

#	Component	MPN	Supplier	Unit	Total (110)
1	ESP32-S3-MINI-1U	ESP32-S3-MINI-1U-N8	Mouser	\$3.80	\$418
2	OLED 1.0" SSD1327	ER-OLED013-1W	BuyDisplay	\$5.50	\$605
3	Camera HM01B0	HM01B0-AAA	ArduCam	\$4.20	\$462
4	Weather BME280	BME280	Mouser 828-BME280	\$2.80	\$308
5	IMU ICM-42688-P	ICM-42688-P	Mouser	\$3.50	\$385
6	Light APDS-9960	APDS-9960	Mouser 630-APDS-9960	\$2.10	\$231
7	Qi Rx IC BQ51013B	BQ51013BRHLR	Mouser 595-BQ51013BRHLR	\$2.80	\$308
8	Qi Rx coil 30mm	WE 760308101	Mouser / Alibaba	\$1.50	\$165
9	PMIC BQ25892	BQ25892RTWR	Mouser	\$2.60	\$286
10	LiPo 280mAh	Custom	Grepow	\$6.50	\$715
11	Copy button	EVQ-Q2C03W	Mouser	\$0.25	\$28
12	RGB LED	APTR3216ZGCK	Mouser	\$0.20	\$22
13	LDO 1.8V	AP2112K-1.8TRG1	DigiKey	\$0.30	\$33
14	Passives	Various	Mouser	\$3.00	\$330
15	PCB, 4-layer	Custom	PCBWay	\$3.50	\$385
16	SMT assembly	Turnkey	PCBWay	\$12.00	\$1,320
17	SS enclosure, 2-part CNC	Custom	PCBWay CNC	\$40.00	\$4,400
18	Qi Tx pad (charger)	OEM custom	Alibaba	\$9.00	\$900
19	Packaging	Custom	Alibaba	\$4.00	\$400

Grand total, 100-unit run + 15% contingency: ~\$12,340

Roadmap

Phase	Status	Scope
0 — Design	COMPLETE	Device spec, BOM, firmware architecture, API design, manufact
1 — Engineering	NOT STARTED	PCB schematic (KiCad 8.0), PCB layout, enclosure CAD, Gerber
2 — Prototype	NOT STARTED	10-unit order, firmware v0.1–v0.3, hardware validation
3 — Production	NOT STARTED	100-unit order, factory flash + provisioning, QA, packaging
4 — Launch	NOT STARTED	My Devices page, Log tab display, OTA infra, FCC/CE, retail at

The Interface

The LOT AI sends a short message — “Coffee time!” — to the device. It appears on the mirror-finished square on the desk. The Copy button closes the loop: one press logs the moment (temperature, humidity, light, orientation, optional camera frame) back to lot-systems.com/log, tagged [COSMO® Cube]. No screen feed, no scroll — one notification, one button, one record.

Full specification: docs/hardware/COSMO-DEVICE-SPEC-v1.md · Firmware: COSMO-FIRMWARE-v1.md · API: COSMO-SOFTWARE-API-v1.md · Manufacturing: COSMO-MANUFACTURING-v1.md · Charger: COSMO-CHARGER-SPEC-v1.md · Continuity note: COSMO-HARDWARE-CONTINUITY-2026-09-03.md